CS 471– Lab 1

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Part 2: Task 2:

Step 2:

397	46.933928500	2a02:ce0:1002:f005:…	2a02:ce0:1:28::5633…	TCP	86 27581 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
398	46.933928500	192.168.1.112	86.51.94.80	TCP	66 27580 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
399	46.945449700	86.51.94.80	192.168.1.112	TCP	66 80 → 27580 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1400 SACK_PERM WS=128
400	46.945611800	192.168.1.112	86.51.94.80	TCP	54 27580 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0

Step 3:

Wireshark · Follow TCP Stream (tcp.stream eq 24) · Wi-Fi

```
GET /connecttest.txt HTTP/1.1
Connection: Close
User-Agent: Microsoft NCSI
Host: www.msftconnecttest.com


HTTP/1.1 200 OK
Content-Length: 22
Date: Fri, 27 Feb 2026 00:37:55 GMT
Connection: close
Content-Type: text/plain
Cache-Control: max-age=30, must-revalidate

Microsoft Connect Test
```

Part 4:

Task 1:

	TCP or UDP	Reasons
Reliability and Connection Establishment	TCP	TCP is connection-oriented and guarantees reliable delivery. UDP is connectionless and does not guarantee delivery.
Data Integrity and Ordering	TCP	TCP ensures data arrives in order and without errors. UDP does not guarantee order or recovery of lost packets.

Task 2:

	TCP	UDP
Use cases	Web browsing (HTTP/HTTPS), File transfer (FTP) Email	Online gaming Video streaming Voice calls
Performance	Slower. Reliable and accurate	Faster, No guarantee of Delivery